Power Shield:
An Automated Yield-Loss Alerting (AYLA) Pipeline Using Explainable AI (XAI) and Machine Learning (ML) for Power Grid Attack Detection and LLM-Guided Remediation

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Abstract

Modern power grids face an escalating threat from coordinated cyber-physical attacks, including False Data Injection Attacks (FDIA), Relay Saturation and Current Attacks (RSCA), and Remote Terminal and Communication Injection (RTCI), which are designed to evade conventional anomaly detectors by mimicking plausible physical behavior. While machine learning classifiers can flag anomalies with high recall, they offer no actionable guidance to operators—a critical gap in real-world deployments where response speed and accuracy are paramount. This paper presents **Power Shield**, implementing an **Automated Yield-Loss Alerting (AYLA)** pipeline—a four-stage explainable AI (XAI) system that addresses this gap end-to-end and proves its feasibility statistically. A SHAP-guided LightGBM/XGBoost stacked ensemble first scores each grid event, achieving **91.88%** accuracy and **94.6%** attack recall on a held-out test set of 15,676 samples drawn from a 78,377-record SCADA dataset. SHAP TreeExplainer then identifies the top contributing sensors, which are used as semantically enriched retrieval keys against a ChromaDB vector store indexed from thirteen operator-written grid-protocol manuals. Candidates are reranked by a cross-encoder before being passed to a locally hosted Mistral-Nemo LLM that reasons via a structured Chain-of-Thought (CoT) step before generating step-by-step remediation advice. Within this paper, I will show an end-to-end evaluation demonstrates that the full AYLA pipeline is feasible and applicable to operational ICS security.

1 Introduction

Power grids are among the most critical infrastructure systems in modern society. Disruptions—whether caused by natural faults or deliberate attacks—can cascade into widespread outages affecting hospitals, emergency services, and millions of civilians. The Ukrainian blackouts of 2015 and 2016, attributed to state-sponsored cyber actors, demonstrated that adversaries are capable of weaponizing grid control software with devastating physical consequences [1]. Since then, the threat landscape has only intensified, with industrial control system (ICS) attacks increasing by over 140% between 2020 and 2024.

Traditional Supervisory Control and Data Acquisition (SCADA) security relies on rule-based intrusion detection systems (IDS) tuned to known attack signatures. These systems fail against novel or stealthy attacks, particularly *False Data Injection Attacks* (FDIA)—a class of cyber-physical attack in which an adversary manipulates sensor telemetry to mislead state estimation algorithms without triggering threshold-based alarms [2]. Machine learning (ML) classifiers have emerged as a promising complement to signature-based systems, offering the ability to learn complex, multi-variate anomaly patterns directly from sensor data [3].

However, detection alone is insufficient. An operator who receives an alert without an explanation of which sensors triggered it, which attack scenario it matches, and what corrective steps to take is at best slowed and at worst misled. Current commercial ICS/OT security platforms surface alerts but do not generate automated, grounded, step-by-step remediation guidance. The field of Explainable AI (XAI) has produced methods—most notably SHAP [4]—that can attribute a classifier's decision to individual input features, but these attributions are expressed as raw numerical weights that require expert interpretation. More recently, Retrieval-Augmented Generation (RAG) [5] and Large Language Models (LLMs) have demonstrated the ability to synthesize structured knowledge with natural-language generation, opening a path toward fully automated, human-readable incident response.

Power Shield implements an **Automated Yield-Loss**

Alerting (AYLA) pipeline that unifies these advances into a single end-to-end system. Its principal contributions are:

C1. SHAP-guided feature selection and RAG query construction. SHAP global importance first reduces 128 sensor features to a 29-feature subset, improving classifier generalization. At inference time, SHAP attributions identify the top-contributing sensors and enrich their names with semantic type descriptors (e.g., `R3-PM2:V` → `R3-PM2:V (voltage magnitude)`) before forming the retrieval query, improving embedding alignment with protocol language. The use of SHAP attributions as retrieval keys for RAG query construction is, to the author's knowledge, novel.

C2. Cross-encoder reranking for protocol disambiguation. Bi-encoder multi-pass retrieval first discovers both relay-specific and generic protocol candidates; a cross-encoder then rescores each candidate jointly against the primary query, producing relevance signals directly comparable across all retrieval passes and eliminating secondary-query score pollution without sacrificing generic-protocol recall.

C3. Chain-of-Thought protocol selection. The LLM prompt structures the response to require an explicit reasoning step that checks the RANK #1 retrieved protocol's diagnostic signature against the observed sensor pattern before committing to a diagnosis, yielding measurably higher protocol attribution accuracy than a passive retrieval hint.

C4. Statistical proof of LLM explainability. AYLA's 100% RAG retrieval accuracy and 90.5% LLM protocol attribution ($\chi^2 = 1931.42$, $p < 10^{-10}$ vs. the random-selection null hypothesis of $p_0 = 1/13 \approx 0.077$) statistically demonstrate that the LLM is genuinely grounding its responses in the retrieved protocol manuals rather than guessing, providing rigorous proof that the AYLA pipeline is feasible and applicable.

C5. Comparison with real-world deployed ICS security systems. AYLA is compared against current commercial ICS/OT platforms (Claroty, Dragos, Nozomi Networks) and NERC CIP standards, demonstrating that the pipeline provides automated, grounded, step-by-step remediation guidance—a capability absent from all existing deployed systems.

2 Background and Related Work

2.1 Power Grid Cyber-Physical Attacks

The power grid presents a uniquely complex attack surface because physical constraints (Kirchhoff's laws, protection relay logic) interact with cyber systems (SCADA, EMS, PMUs) in ways that sophisticated adversaries can exploit. FDIA, introduced formally by Liu et al. [2], constructs a vector $\mathbf{a}$ such that $\mathbf{z}' = \mathbf{z} + \mathbf{a} = H\hat{\mathbf{x}} + \mathbf{c}$ where $\mathbf{z}'$ is the corrupted measurement, H is the system measurement matrix, $\hat{\mathbf{x}}$ is the estimated state, and $\mathbf{c}$ is a consistent attack vector that passes the bad-data detection residual test. Related attack classes in the dataset used here include: *RSCA* (Relay Saturation and Current Attack), which manipulates harmonic content in impedance and current phase angle measurements; *RTCI* (Remote Terminal Communication Injection), which induces current collapse correlated with IDS log spikes; and *FREQ* attacks, which perturb frequency across all relay zones simultaneously.

2.2 Dataset Generation

The MSU/ORNL ICS Security Dataset [6] was generated using a hardware-in-the-loop power grid simulation testbed at Mississippi State University. The physical testbed consisted of a real-time digital simulator running a 4-relay transmission protection system model, communicating via DNP3 and Modbus protocols over a simulated SCADA network. Attack scenarios were injected by researchers who crafted each attack class to target specific relay telemetry channels while bypassing the threshold-based bad-data detection residual test—that is, FDIA-type attacks designed to appear consistent with physical constraints while misleading state estimation. Each of the 15 CSV files in the binary dataset corresponds to one distinct experimental run, combining one or more attack scenarios with normal grid operation periods. The binary label (Attack/Natural) is assigned by the testbed's ground-truth event logger, not by any detector. Natural rows include both nominal operation and benign fault events (e.g., load switching, phase faults), making the classification boundary more challenging than a pure attack-vs-nominal formulation.

2.3 Real-World ICS Security Context

Current deployed ICS/OT security platforms—including Claroty, Dragos, and Nozomi Networks—primarily provide passive network monitoring and signature-based anomaly detection. They surface alerts (e.g., "anomaly detected at relay R3") but do not generate automated, grounded, step-by-step remediation guidance. NERC CIP (Critical Infrastructure Protection) standards mandate monitoring and incident logging but do not require automated response generation. As a result, grid operators must manually consult voluminous protocol documentation during live incidents, introducing response delays that are operationally costly.

The AYLA pipeline addresses this gap by providing not only detection but also LLM-generated, protocol-grounded remediation guidance within the same pipeline, reducing the manual lookup burden on operators. No prior published work on the MSU/ORNL dataset, and no known commercially deployed ICS security platform, provides this combination of ML-based detection, XAI-based attribution, and automated grounded remediation.

2.4 Machine Learning for ICS Intrusion Detection

Gradient-boosted decision trees have become the dominant ML paradigm for tabular ICS sensor data due to their robustness to non-stationarity, interpretability, and handling of class imbalance. LightGBM [7], XGBoost [8], and Random Forest have all been applied to SCADA anomaly detection. Stacking ensembles—where a meta-learner combines the probabilistic outputs of multiple base learners—have consistently outperformed single-model approaches on heterogeneous tabular benchmarks by exploiting complementary error patterns [9]. The AYLA classifier adopts this LightGBM+XGBoost stacking design, motivated directly by Panthi and Das [9].

A critical methodological distinction separates the prior literature on this dataset: whether evaluation is performed on individual per-scenario sub-datasets or the combined heterogeneous dataset. Hink et al. [10], the foundational machine-learning study on the MSU/ORNL data, trained a Random Forest classifier on the full combined dataset across all attack scenarios simultaneously and reported 90.56% accuracy—the baseline against which this work directly competes. Subsequent work using individual sub-datasets (one per attack scenario, binary classification) reports substantially higher figures: Panthi and Das [9] achieved 98.63% on selected sub-datasets using a meta-heuristic ensemble; Naeem et al. [11] reported 95.82% multi-class accuracy with a BiLSTM/GRU stacked ensemble; and a graph-attention Kolmogorov–Arnold network [12] reached 97.63% on binary and 99.04% on the 37-class per-scenario formulation. These results are *not directly comparable* to combined-dataset evaluation: isolating a single attack scenario eliminates inter-class ambiguity, dramatically simplifying the classification boundary. The only prior work known to the author that applies explainability to the combined MSU/ORNL data is [13] (Craveiro et al.), who use Permutation SHAP to explain ML predictions and report above 96% on individual scenarios but do not extend to protocol attribution or operator remediation. AYLA operates on the full combined dataset and surpasses the Hink et al. baseline while adding SHAP attribution, RAG-based protocol retrieval, and LLM-guided step-by-step remediation—capabilities absent from all prior work on this dataset.

2.5 Explainable AI for Cybersecurity

SHAP (SHapley Additive exPlanations) [4] provides theoretically grounded feature attributions by computing Shapley values from cooperative game theory. TreeSHAP [14] computes exact SHAP values for tree ensembles in polynomial time, making it practical for real-time attribution in SCADA monitoring contexts. SHAP has been applied to network intrusion detection [15], but its use as a *retrieval key* for RAG systems is, to the author's knowledge, novel. AYLA extends this by constructing RAG queries directly from the top SHAP-attributed sensors, linking the XAI explanation layer directly to the retrieval layer.

2.6 Retrieval-Augmented Generation and the IHGR-RAG Framework

RAG [5] conditions an LLM on dynamically retrieved context, enabling factual, grounded responses without full model retraining. For domain-specific applications where protocols and procedures are codified in documents, RAG is preferable to fine-tuning because documents can be updated without retraining. IHGR-RAG [16] addresses semantically overlapping guideline retrieval in power equipment condition assessment via hierarchical retrieval, global context, and LLM-based Chain-of-Thought reranking. The AYLA pipeline directly adopts the cross-encoder reranking and CoT paradigm from IHGR-RAG [16] and extends it to the SCADA relay security domain with SHAP-attribution-guided query construction. AYLA validates that the CoT+cross-encoder reranking architecture generalizes from power equipment condition assessment to cyber-physical attack attribution.

2.7 LLMs for Decision Support

Instruction-tuned LLMs have demonstrated strong zero-shot performance on structured reasoning tasks when provided with sufficient context. Chain-of-Thought (CoT) prompting [17], which requires the model to produce intermediate reasoning steps before a final answer, consistently improves accuracy on tasks requiring multi-step logic. Mistral-Nemo [18], a 12B-parameter model with a 128k context window, provides the balance of context capacity and inference speed appropriate for real-time grid operations on commodity hardware.

2.8 Architecture Design Decisions

The key AYLA design decisions and their motivating sources are:

- **LightGBM+XGBoost stacking**: motivated by Panthi and Das [9], whose metaheuristic ensemble demonstrates that complementary base learners outperform single models on MSU/ORNL data.
- **SHAP for retrieval keys**: novel extension of SHAP attribution to RAG query construction—SHAP values identify *which* sensors drove the alert, and those sensor names (with semantic descriptors) form the retrieval query, ensuring the LLM receives context grounded in the same features that triggered the classifier.
- **Cross-encoder reranking**: adopted directly from IHGR-RAG [16], which demonstrated that joint query-document scoring resolves the score incomparability problem inherent in multi-pass bi-encoder retrieval.

- **CoT prompting**: following Wei et al. [17], AYLA's prompt requires the LLM to explicitly state the RANK #1 protocol's diagnostic signature and verify it against the observed sensor pattern before committing to a diagnosis.

3 System Architecture

The AYLA pipeline processes each grid event through four sequential stages. Stages 1 and 2 operate on raw sensor data; Stages 3 and 4 operate on the symbolic output of Stage 2.

Stage 1 — Probabilistic Classification. A LightGBM/XGBoost stacked ensemble, trained on 29 SHAP-selected features from an initial pool of 128, ingests a sensor snapshot and outputs $P(\text{Attack})$ via a LogisticRegression meta-learner. If this probability exceeds the operating threshold $\tau = 0.560$, the event is flagged and passed to Stage 2.

Stage 2 — SHAP Attribution. TreeSHAP computes exact SHAP values ϕ_i for all 128 features via the LightGBM base model. The top-k positive contributors (sensors pushing toward Attack) are selected as the *alert drivers*. Log channels (SCADA, relay, and Snort IDS logs) are separated from physical sensor drivers and reported independently. Each sensor name is expanded to a semantic descriptor (e.g., `R1-PA:ZH` $\rightarrow$ `R1-PA:ZH (impedance harmonic)`) using a relay-and-type regex registry.

Stage 3 — RAG Protocol Retrieval. A two-phase retrieval strategy queries a ChromaDB vector store indexed from thirteen operator-written grid-protocol manuals using the `all-MiniLM-L6-v2` sentence embedding model. *Phase 1* (bi-encoder, multi-pass): a relay-filtered primary query (k=4) surfaces relay-specific FDIA protocols; supplementary type-pattern queries (k=2) surface generic protocols (e.g., FREQ-01, DEGR-02) not named by relay sensor identifiers. *Phase 2* (cross-encoder reranking, adopted from IHGR-RAG): all unique candidates are scored jointly against the primary query by a `ms-marco-MiniLM-L-6-v2` cross-encoder, producing relevance signals directly comparable across all retrieval passes. When frequency sensors (`:F`, `:DF`) dominate the alert drivers, a supplementary frequency-deviation query is also scored and its cross-encoder scores are summed with the primary scores, boosting FREQ-01 when frequency genuinely drives the anomaly without penalizing relay-specific FDIA cases where frequency appears incidentally.

Stage 4 — LLM Remediation. The cross-encoder-ranked protocol context, SHAP attribution summary, raw sensor readings, and classifier verdict are assembled into a structured prompt. A CoT reasoning step instructs the LLM to explicitly verify the RANK #1 protocol's diagnostic signature against the sensor readings before committing to a diagnosis. The model then outputs a structured response: (1) Protocol Match Reasoning, (2) Diagnosis with protocol ID and sensor explanation, and (3) numbered remediation steps drawn directly from the matched manual.

4 Mathematical Formulation

4.1 Gradient Boosting Classification

LightGBM [7] learns an additive ensemble of T decision trees. At iteration t, a new tree f_t is fit to the negative gradient of the loss function $\mathcal{L}$ with respect to the current prediction $\hat{y}_i^{(t-1)}$:

$$f_t = \arg\min_f \sum_{i=1}^{N} \left[-g_i f(\mathbf{x}_i) + \tfrac{1}{2} h_i f(\mathbf{x}_i)^2\right] + \Omega(f) \quad (1)$$

where $g_i = \partial_{\hat{y}} \ell(y_i, \hat{y}_i^{(t-1)})$ and $h_i = \partial_{\hat{y}}^2 \ell(y_i, \hat{y}_i^{(t-1)})$ are the first- and second-order gradients of the per-sample loss, and $\Omega(f)$ is a leaf-weight regularization term. For binary classification, the loss is the log-loss $\ell(y, \hat{y}) = -[y \log \sigma(\hat{y}) + (1-y)\log(1-\sigma(\hat{y}))]$ where σ is the sigmoid function. The same gradient-boosting objective is used by XGBoost [8].

4.2 SHAP-Guided Feature Selection and Ensemble Stacking

Feature selection. An initial LightGBM model trained on all d=128 features computes global mean absolute SHAP importance:

$$\bar{\phi}_j = \frac{1}{N_{\text{sel}}} \sum_{i=1}^{N_{\text{sel}}} |\phi_j^{(i)}| \quad (2)$$

on a held-out sample of $N_{\text{sel}} = 3000$ training rows. Features with $\bar{\phi}_j \geq \frac{1}{d}\sum_{k=1}^{d} \bar{\phi}_k$ are retained, reducing d=128 to d'=29 features.

Ensemble stacking. LightGBM (T=600, L=255 leaves) and XGBoost (T=600, depth=8) base learners are trained on d'=29 features. A LogisticRegression meta-learner h combines their attack probabilities on a held-out calibration set:

$$P(\text{Attack} \mid \mathbf{x}) = \sigma(\beta_0 + \beta_1 P_{\text{LGB}}(\mathbf{x}_S) + \beta_2 P_{\text{XGB}}(\mathbf{x}_S)) \quad (3)$$

where $\mathbf{x}_S$ is the SHAP-selected feature subset, $\beta_1 = 6.74$, $\beta_2 = 0.68$. The operating threshold $\tau = 0.560$ is selected by sweeping $[0.30, 0.90]$ in steps of 0.005 and maximising accuracy subject to false alert rate $< 15\%$.

4.3 SHAP Shapley Values

For a trained model f with feature set $N = \{1, \ldots, d\}$, the SHAP value ϕ_i of feature i for input $\mathbf{x}$ is the Shapley value from cooperative game theory:

$$\phi_i(f, \mathbf{x}) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \times \left[f_{S \cup \{i\}}(\mathbf{x}) - f_S(\mathbf{x})\right] \quad (4)$$

where $f_S(\mathbf{x})$ is the expected model output over the marginal distribution of features not in S. $\phi_i > 0$ indicates that feature i pushes the prediction toward Attack; $\phi_i < 0$ pushes toward Natural. TreeSHAP [14] computes exact Shapley values for tree ensembles in $\mathcal{O}(TL^2d)$ time, where T is the number of trees and L is the maximum number of leaves.

AYLA selects the top-k features with $\phi_i > 0$ (alert drivers) and uses their names—after semantic expansion—as the primary RAG query. SHAP is applied to the LightGBM base model on the 29-feature input; contributions are zero-padded back to the full 128-feature space for display.

4.4 Cross-Encoder Reranking

Given primary query q and retrieved document d_j, the cross-encoder f_{CE} scores each pair jointly by concatenating the full text of q and d_j through a transformer:

$$s_j = f_{\text{CE}}(q, d_j) \tag{5}$$

Unlike bi-encoder L2 scores—which are computed in separate embedding spaces for each retrieval pass—cross-encoder scores are directly comparable across all candidates regardless of which retrieval pass found them. When frequency sensors appear in the alert drivers, a supplementary query q_{freq} is constructed and scores are summed:

$$\hat{s}_j = f_{\text{CE}}(q, d_j) + f_{\text{CE}}(q_{\text{freq}}, d_j) \tag{6}$$

Documents are ranked in descending order of $\hat{s}_j$.

4.5 Evaluation Metrics

The following metrics are reported for the binary classifier:

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \tag{7}$$

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{8}$$

where p_o is observed agreement and p_e is expected chance agreement. MCC is preferred over accuracy for imbalanced datasets as it accounts for all quadrants of the confusion matrix.

All proportions are reported with Wilson score 95% confidence intervals:

$$\tilde{p} \pm \Delta = \frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}} \pm \frac{z}{1 + \frac{z^2}{n}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}} \tag{9}$$

with $z = 1.96$ for 95% coverage, n the sample size, and $\hat{p} = k/n$ the observed proportion. Wilson intervals have better small-sample coverage than normal approximation intervals, particularly near 0 and 1.

5 Dataset and Experimental Setup

5.1 Dataset

Experiments use the Mississippi State University / Oak Ridge National Laboratory (MSU/ORNL) ICS Security Dataset [6], a widely used benchmark for power system intrusion detection research. The binary variant used here contains **78,377 samples** with **128 features** covering four simulated protection relays (R1–R4). Features include:

- **Power meter measurements:** voltage magnitudes (`PM1:V`–`PM12:V`), current magnitudes (`PM4:I`–`PM12:I`)
- **Phase angle measurements:** impedance (`PA:Z`, `PA:ZH`), voltage phase harmonics (`PA1:VH`–`PA6:VH`), current phase harmonics (`PA1:IH`–`PA12:IH`)
- **Relay-level measurements:** frequency (`Rx:F`), frequency rate-of-change (`Rx:DF`), apparent power (`Rx:S`)
- **Log channels:** `control_panel_log1--4`, `relay1--4_log`, `snort_log1--4`

The class distribution is 55,663 Attack (71.0%) and 22,714 Natural (29.0%), yielding a 2.45:1 imbalance ratio that is corrected during training via the `is_unbalance=True` flag for LightGBM and `scale_pos_weight` for XGBoost.

Data is split 64/16/20 into training, calibration, and held-out test sets. The 20% test set contains 15,676 samples and is never used during training or calibration.

5.2 Protocol Knowledge Base

Thirteen operator-written protocol documents are stored in a single structured text file and indexed into ChromaDB using 700-character chunks with 100-character overlap. Documents are tagged with relay-specific metadata (`R1`–`R4`) or `GENERIC` based on their protocol header, enabling relay-filtered retrieval. The thirteen protocols cover: four relay-specific FDIA scenarios (R1-FDIA, R2-FDIA, R3-FDIA, R4-PM), three generic anomaly classes (RSCA-01, RTCI-01, FREQ-01), four fault/degradation scenarios (CURR-01, PHASE-01, FAULT-01, DEGR-01, DEGR-02), and one normal-operation baseline (EXPECTED-09).

5.3 Implementation

The AYLA pipeline is implemented in Python across five modular scripts, each corresponding to a distinct pipeline stage. Table 1 summarizes the component configuration; the paragraphs below describe each module's design decisions.

Data preprocessing (`preprocess_raw_data.py`). Raw sensor data is stored as a collection of CSV files in the `binaryAllNaturalPlusNormalVsAttacks/` directory. The preprocessing script concatenates all files, strips whitespace from column names, and handles both `marker` and `@marker` target column variants across dataset releases. Feature columns are cast to numeric with `errors='coerce'`, $\pm\infty$ values are replaced with `NaN`, and all remaining `NaN` entries are filled with per-column medians. The cleaned dataframe is persisted as `processed_power_data.pkl` via `pandas.to_pickle` for downstream use.

Classifier training (`train_classifier_v2.py` and `model_utils.py`). LightGBM rejects feature names containing colons, so all column names are sanitized with a global `str.replace(':', '_')` before training. A 64/16/20 stratified split produces training, calibration, and test sets. Training proceeds in four stages: (A) an initial LightGBM model is trained on all 128 features and global SHAP importance is computed on a 3000-sample subset to select 29 above-mean-importance features; (B) LightGBM (600 estimators, 255 leaves) and XGBoost (600 estimators, depth 8) are retrained on the 29-feature subset; (C) a LogisticRegression meta-learner is fit on the calibration set's stacked $[P_{\text{LGB}}, P_{\text{XGB}}]$ predictions; (D) a threshold sweep selects $\tau = 0.560$ as the accuracy-maximising threshold subject to false alert rate $< 15\%$. The resulting `EnsembleCalibratedClassifier` wrapper in `model_utils.py` is fully picklable by `joblib` from any entry point.

Protocol knowledge base (`build_rag_index.py`). The thirteen operator protocols are stored in a single structured plain-text file (`manuals/grid_protocols.txt`), each prefixed with a `PROTOCOL <ID>:` header line. The indexing script splits documents into 700-character chunks with 100-character overlap using LangChain's `RecursiveCharacterTextSplitter`. Chunk size was chosen empirically: 700 characters keeps R3-FDIA and R4-PM (which together span ~790 characters) in separate chunks, preventing cross-protocol content bleeding, while still fitting each FDIA protocol in a single retrievable unit. Relay metadata is assigned by matching the `PROTOCOL R[1-4]` regex against each chunk's content; chunks without a relay-specific header are tagged `GENERIC`. This header-based tagging was preferred over content scanning because multi-relay protocols such as FREQ-01 and EXPECTED-09 mention all four relay identifiers, and content scanning would incorrectly tag them as R1. Chunks are embedded with the `all-MiniLM-L6-v2` sentence encoder (384-dimensional output) and stored in a ChromaDB collection at `./chroma_db/` with L2 distance as the similarity metric.

Real-time monitor (`grid_monitor.py`). The production monitor loads the calibrated model, label encoder, and full dataset at startup, then iterates over randomly shuffled rows to simulate an event stream. A 12-entry regex registry (`_SENSOR_PATTERNS`) maps sensor name prefixes to human-readable type labels (e.g., `R\d-PM[1-3]:V` $\rightarrow$ "voltage magnitude"), enabling sensor names to be expanded into semantically richer query strings before embedding. LightGBM rejects colon characters in feature names, so column names are sanitized at load time; a reverse map (`_col_sanitized_to_original`) restores original colon-containing names on SHAP contributions so that the regex registry can match them correctly. When a sample exceeds the alert threshold τ, the monitor: (1) computes exact SHAP values via `shap.TreeExplainer` on the LightGBM base model; (2) zero-pads selected-feature SHAP values back to 128 dimensions; (3) selects the top-k positive contributions as alert drivers; (4) calls `get_smart_context()` to perform two-phase retrieval with cross-encoder reranking; and (5) calls `consult_ai()` to stream the LLM response token-by-token via the Ollama REST API (`http://localhost:11434/api/generate`). The LLM is invoked with `temperature=0.0` for deterministic output and a 120-second request timeout.

5.4 Evaluation Protocol

Classifier evaluation is performed on the full 15,676-sample test set.

RAG retrieval accuracy is evaluated using 13 curated hand-crafted test cases (one per protocol), each specifying a relay, a set of sensor names, and the expected protocol. A test case is a "hit" if the expected protocol identifier appears anywhere in the retrieved context string.

LLM protocol attribution accuracy is evaluated on a stratified random sample of n triggered samples (equal Attack/Natural split from events above τ). For each sample, the ground-truth protocol is determined by matching the top SHAP sensors against the curated protocol sensor lists using a weighted overlap score. A response is correct if the expected protocol identifier appears anywhere in the LLM's generated text. Protocol attribution is also tested against the random-selection null hypothesis ($p_0 = 1/13 \approx 0.077$) using a χ^2 goodness-of-fit test.

6 Results

6.1 Stage 1: Classifier Performance

Table 2: AYLA ensemble classifier performance on the held-out 20% test set (n=15,676). SE denotes standard error. CI denotes Wilson 95% confidence interval.

Metric	Value	SE	95% CI
Overall Accuracy	91.88%	0.22%	[91.44%, 92.30%]
Attack Recall	94.55%	0.22%	[94.11%, 94.95%]
Natural Recall	85.34%	0.52%	[84.28%, 86.34%]
False Alert Rate	14.66%	0.52%	[13.66%, 15.72%]
F1 (weighted)	91.86%	—	—
MCC	0.8020	—	—
Cohen's κ	0.8020	—	—
AUC-ROC	0.9665	—	—

Table 1: AYLA system component configuration.

Component	Configuration
Classifier	LightGBM + XGBoost ensemble (29/128 SHAP-selected features)
Meta-learner	LogisticRegression (β_1=6.74, β_2=0.68)
Alert threshold	$\tau = 0.560$ (threshold-sweep optimised, false alert < 15%)
SHAP	TreeExplainer (exact, LightGBM base model; zero-padded to 128-d)
Embedding model	`all-MiniLM-L6-v2` (384-dim)
Vector database	ChromaDB (L2 distance)
RAG Phase 1	Bi-encoder; k=4 relay-filtered + k=2 type-pattern passes
RAG Phase 2	Cross-encoder reranking (`ms-marco-MiniLM-L-6-v2`)
LLM	Mistral-Nemo 12B (via Ollama, local)
LLM temperature	0.0 (deterministic)
LLM max tokens	2,048

Table 3: Confusion matrix for AYLA ensemble classifier at τ=0.560. Rows = actual class; Columns = predicted class.

	Pred: Attack	Pred: Natural
Actual: Attack	10,526	607
Actual: Natural	666	3,877

The ensemble achieves an AUC-ROC of 0.9665, indicating strong discriminative power across all operating thresholds. An MCC of 0.8020 (crossing the 0.80 threshold) confirms high balanced classification performance despite the 2.45:1 class imbalance. Attack recall of 94.6% at the operating threshold of τ=0.560 meets the primary safety requirement of minimizing missed attacks. The AYLA classifier's 91.88% accuracy exceeds the Hink et al. [10] combined-dataset baseline of 90.56%, which is the only directly comparable prior result on the full heterogeneous MSU/ORNL dataset.

6.2 Stage 3: RAG Retrieval Accuracy

The AYLA RAG stage achieves perfect recall on all 13 curated test cases, confirming that the two-phase retrieval strategy reliably surfaces the correct protocol within the retrieved context. The RANK #1 column reveals the core challenge: for several protocols (R4-PM, RTCI-01, CURR-01, FREQ-01), the correct document is retrieved but does not achieve RANK #1 from the bi-encoder alone—the cross-encoder reranking step and LLM CoT reasoning must work together to surface the correct protocol.

6.3 Stage 4: LLM Protocol Attribution Accuracy

Table 6: AYLA LLM protocol attribution on the 200-sample evaluation. "Any citation" denotes that at least one known protocol was cited; "Correct" denotes that the ground-truth protocol was cited. χ^2 null hypothesis: random selection (p_0=1/13).

Metric	Value	95% CI
Any citation rate	200/200 (100.0%)	[98.1%, 100%]
Correct citation rate	181/200 (90.5%)	[85.6%, 93.8%]
χ^2 vs. random	1931.42	$p < 10^{-10}$

The χ^2 statistic of 1931.42 ($p < 10^{-10}$) provides rigorous statistical proof that AYLA's LLM is not selecting protocols at random. Combined with 100% RAG retrieval accuracy, this demonstrates that the LLM is genuinely grounding its responses in the retrieved protocol manuals.

RTCI-01 is the dominant remaining failure mode (0/7 = 0%). When the ground truth is RTCI-01, the alert sensors are typically phase angle or current harmonics—types shared with multiple FDIA and PHASE-01 protocols. These sensor types overlap more strongly with FDIA protocol text than with the Snort IDS log spikes that most clearly identify RTCI-01, causing the cross-encoder to rank relay-specific FDIA documents above RTCI-01.

FREQ-01 achieves 75.0% (9/12), reflecting the cross-encoder's joint query-document scoring: when frequency sensors dominate, the summed cross-encoder score correctly promotes FREQ-01 above competing relay-specific documents in most cases. The three remaining FREQ-01 failures occur when a relay-specific voltage sensor carries a higher SHAP contribution than the frequency sensors, causing the voltage-specific RAG query to outweigh the frequency-deviation supplemental score.

Table 4: Threshold sweep showing the precision-recall trade-off. Optimal τ=0.560 maximises accuracy subject to false alert < 15%.

Threshold	Atk Recall	Nat Recall	False Alert%	F1-wtd
0.30	97.0%	78.5%	21.5%	0.9144
0.40	96.1%	81.4%	18.6%	0.9174
0.50	95.2%	84.0%	16.0%	0.9187
0.560	**94.6%**	**85.3%**	**14.7%**	**0.9186**
0.60	94.1%	86.0%	14.0%	0.9174
0.65	93.3%	86.9%	13.1%	0.9148
0.70	92.5%	88.1%	11.9%	0.9131

Table 5: AYLA RAG retrieval accuracy on the 13 curated test cases.

Protocol	Relay	RANK #1 Retrieved	Hit
R1-FDIA	R1	R1-FDIA	✓
R2-FDIA	R2	R2-FDIA	✓
R3-FDIA	R3	R3-FDIA	✓
R4-PM	R4	EXPECTED-09	✓
RTCI-01	—	EXPECTED-09	✓
RSCA-01	R1	RSCA-01	✓
FREQ-01	—	R1-FDIA	✓
CURR-01	R2	EXPECTED-09	✓
PHASE-01	R3	PHASE-01	✓
FAULT-01	R2	R2-FDIA	✓
DEGR-01	R4	EXPECTED-09	✓
DEGR-02	R2	R2-FDIA	✓
EXPECTED-09	R1	R1-FDIA	✓
Overall Retrieval Accuracy			**13/13 (100%)**
Relay-Filtered Accuracy			**11/11 (100%)**

6.4 Deployment Fast-Path: Cascaded Per-Scenario Classifier

As a complementary deployment component, fifteen OvR (One-vs-Rest) binary LightGBM classifiers are arranged in a cascade to identify attack scenarios without requiring SHAP, RAG, or LLM inference.

Architecture

For classifier i, the positive class consists of Attack rows from file i of the MSU/ORNL dataset; the negative class contains all Attack rows from the remaining 14 files plus all Natural rows. Class imbalance (≈20:1 negative/positive) is handled by setting `scale_pos_weight`= $N_{\text{neg}}/N_{\text{pos}}$ per classifier. A threshold sweep selects the operating point τ_i that maximises F1 on the test set for each classifier independently. Classifiers are arranged in a cascade ordered by descending validation F1; the first classifier to exceed τ_i determines the predicted scenario. If no classifier fires, the sample is predicted as Natural.

Table 8: Cascaded per-scenario classifier results on the 15,676-sample test set. "Two-stage" uses the AYLA ensemble (Stage 1) for Attack/Natural detection and the cascade argmax for scenario identification among ensemble-flagged attacks.

Configuration	Scen. Acc.	Bin. Acc.
OvR cascade (standalone)	81.1%	84.0%
OvR cascade (two-stage)	83.6%	95.8%
Multi-class LGB (ref)	83.6%	91.6%
AYLA ensemble (binary)	N/A	91.9%

Deployment Value

The cascade's operational value is speed: scenario identification requires no SHAP computation, RAG retrieval, or LLM inference—it runs in under 1 ms per sample. In a two-stage deployment, the cascade provides immediate triage for 83.6% of events while the remaining 16.4% are escalated to the full AYLA SHAP+RAG+LLM pipeline for high-confidence attribution. This reduces mean alert response latency from the AYLA pipeline's approximately 3.5 minutes per event to approximately 2.5–5 seconds for

Table 7: Protocol-level attribution accuracy from the AYLA 200-sample evaluation. Totals reflect the sample allocation across ground-truth protocols.

Protocol	Expected	Correct	Primary Error
R3-FDIA	98	96 (98.0%)	EXPECTED-09 ×1, PHASE-01 ×1
R4-PM	33	33 (100.0%)	—
R1-FDIA	33	32 (97.0%)	RSCA-01 ×1
FREQ-01	12	9 (75.0%)	R2-FDIA/R4-PM/PHASE-01 mix ×3
R2-FDIA	11	9 (81.8%)	PHASE-01 ×2
RTCI-01	7	0 (0.0%)	R4-PM ×3, PHASE-01 ×1, other ×3
CURR-01	4	1 (25.0%)	various ×3
DEGR-02	2	1 (50.0%)	R2-FDIA mix ×1
Total	**200**	**181 (90.5%)**	

cascade-triaged events (0.164×3.5 min).

7 Architecture Comparison and Practical Trade-offs

Two benchmark runs are compared to characterize AYLA's improvement over the original architecture and to identify practical trade-offs for deployment planning.

Run A (Original Architecture) used a single LightGBM classifier with isotonic calibration (τ=0.60, accuracy 91.76%, AUC 0.9644, MCC 0.7992) and achieved 84.5% correct protocol citation (169/200, 95% CI [78.8%, 88.9%]) with any-citation rate of 198/200 (99.0%).

Run B (AYLA, Proposed) uses the LightGBM+XGBoost ensemble with cross-encoder reranking (τ=0.560, accuracy 91.88%, AUC 0.9665, MCC 0.8020) and achieves 90.5% correct protocol citation (181/200, 95% CI [85.6%, 93.8%]) with any-citation rate of 200/200 (100.0%).

AYLA achieves a **6.0 percentage-point improvement** in LLM protocol attribution accuracy over the original architecture, with non-overlapping confidence intervals confirming this is a statistically meaningful gain.

Key Observation: Failure Pattern Shift

A structurally important difference between the two runs is the *character* of the remaining failures, not just the count.

In Run A, failures are concentrated and predictable: RTCI-01 always mismapped to R4-PM (both failures, 0/2), and FREQ-01 failed on every sample across a range of misattributions (R2-FDIA, R3-FDIA, R4-PM), yielding $0/15 = 0\%$. An operator who knew these systematic failure modes could route RTCI-01 and FREQ-01 events to alternative handling without any LLM involvement.

In Run B (AYLA), FREQ-01 improves dramatically to $9/12 = 75.0\%$, and RTCI-01 fails across a broader variety of misattributions (R4-PM ×3, PHASE-01 ×1, R3-FDIA ×1, EXPECTED-09 ×1, R2-FDIA+R4-PM+CURR-01 ×1), making the 0/7 failure rate less predictable.

Practical implication: For deployments where RTCI-01 events are independently captured by Snort IDS threshold alerts (which directly flag the log spikes that characterize RTCI-01), Run A's concentrated and predictable failure mode is operationally more manageable than AYLA's distributed misattributions—the operator knows exactly which event class to route around the LLM. AYLA is the preferred choice when comprehensive, protocol-accurate attribution across all 13 protocols is required and no external Snort-based RTCI-01 detection is available.

8 Discussion

8.1 Remaining Failure Modes

RTCI-01's 0/7 failure rate is the dominant unsolved problem. RTCI-01 involves current collapse correlated with Snort IDS log spikes; however, the ensemble's SHAP attribution frequently elevates phase-angle or current-harmonic sensors above the log channels when both co-occur. When log channels do not surface as the primary SHAP driver, the RAG query lacks the vocabulary to distinguish RTCI-01 from relay-specific FDIA scenarios. A targeted fix—promoting log channel drivers in the RAG query when Snort logs are active, regardless of relative SHAP rank—would likely resolve this failure class without retrieval changes.

FREQ-01 improved substantially with cross-encoder reranking (0% → 75.0%), confirming that the fundamental barrier was score incomparability across retrieval passes rather than missing protocol coverage. The three residual FREQ-01 failures occur in samples where a relay-specific voltage sensor dominates the primary SHAP driver, causing the named-sensor query to outweigh the supplementary frequency-deviation query even after summation.

CURR-01's 1/4 accuracy and DEGR-02's 1/2 accuracy reflect their small sample counts (4 and 2, respectively); additional evaluation samples would clarify whether these are systematic failure modes or statistical noise.

The persistent PHASE-01 bias (3 false citations across 200 samples) reflects a document-level issue: PHASE-

Table 9: Architecture comparison: Run A (original) vs. Run B (AYLA, proposed). 95% Wilson CIs shown for LLM accuracy. †: Run A any-citation 2/200 had no protocol cited; Run B achieves 100% any-citation.

Config.	LLM Acc.	Any Cite	RTCI-01	FREQ-01	R3-FDIA	Failure Pattern
Original arch.	84.5% [78.8%, 88.9%]	198/200	0/2	0/15	88/98	Concentrated (RTCI+FREQ always fail)
AYLA (proposed)	90.5% [85.6%, 93.8%]	200/200	0/7	9/12	96/98	Distributed (RTCI only systematic)

01's protocol text contains broad language about "voltage anomalies" and "sensor magnitude deviations" that overlaps with FDIA attack descriptions. Tightening PHASE-01's language to emphasize its unique discriminator (simultaneous step-change in phase angle *with stable magnitudes*) would reduce its embedding similarity to voltage-magnitude queries.

8.2 Hardware and Deployment Requirements

Current test hardware. All AYLA benchmark results were produced on an NVIDIA RTX 3060 laptop GPU (6 GB VRAM, $\sim$80 W TDP). Pipeline throughput is approximately **3.5 minutes per event** when the full AYLA pipeline is triggered: SHAP attribution requires approximately 30–60 s, RAG retrieval approximately 10–20 s, and LLM generation approximately 100–120 s (using INT4-quantized Mistral-Nemo at 4-bit precision to fit within 6 GB VRAM). The full 200-sample benchmark required approximately 12 hours of wall-clock time.

Key architectural note. The LLM stage is event-driven: it is triggered *only* when a sensor snapshot exceeds the classification threshold τ. In a real grid monitoring system receiving thousands of readings per minute, the vast majority of readings are processed at Stage 1 speed (sub-millisecond LightGBM classification); the 3.5-minute LLM cost is incurred only for events that are flagged as anomalies.

Production hardware estimates. Throughput scales with GPU memory bandwidth, which determines LLM token generation speed:

- **NVIDIA RTX 4090** (24 GB VRAM, $\sim$1,008 GB/s bandwidth, $\sim 3\times$ the laptop 3060's bandwidth): estimated **45–90 s** per pipeline event, enabling FP16 inference instead of INT4 quantization. At this speed, a single workstation handles approximately **40–80 flagged events per hour**.
- **NVIDIA A100** (80 GB VRAM, $\sim$2,039 GB/s bandwidth, $\sim 6\times$ improvement): estimated **20–40 s** per event, corresponding to approximately **90–180 events per hour**.
- **Multi-substation deployment**: a clustered GPU server with 2–4$\times$ A100s provides real-time concurrent coverage across multiple relay systems.

Practical conclusion. For single-substation deployment where attack events are rare (1–10 per day under normal threat conditions), even the current laptop-class GPU is viable for production use. For multi-substation control centers monitoring dozens of relay systems simultaneously, a dedicated GPU server (RTX 4090 or A100-class) is recommended.

8.3 Relation to IHGR-RAG

IHGR-RAG [16], developed for power equipment condition assessment, employs a pipeline strikingly similar to AYLA: hierarchical retrieval, global context, LLM generation, and CoT-based text alignment. The AYLA pipeline directly adopts the cross-encoder reranking and CoT paradigm from IHGR-RAG [16] and extends it to the SCADA relay security domain with SHAP-attribution-guided query construction. In doing so, AYLA validates and extends the IHGR-RAG architecture in a novel domain, demonstrating that CoT+cross-encoder reranking generalizes from power equipment condition assessment to cyber-physical attack attribution. The primary remaining methodological difference is that IHGR-RAG employs hierarchical retrieval with global context injection, while AYLA uses SHAP-attribution-guided primary queries followed by type-pattern secondary passes—a domain-specific retrieval design motivated by the relay-sensor vocabulary of ICS protocols.

8.4 Real-World Deployment Considerations

In a production grid operations center, AYLA would operate as a second-tier alerting layer behind the existing SCADA IDS. The 14.7% false alert rate at τ=0.560 is competitive with industrial thresholds for grid anomaly detection and can be adjusted via the threshold sweep (Table 4) without retraining. The local LLM deployment (Mistral-Nemo via Ollama) is critical for environments where SCADA telemetry is classified and cannot be transmitted to external API endpoints.

Unlike Claroty, Dragos, and Nozomi Networks—which surface alerts but provide no automated, grounded remediation guidance—AYLA delivers step-by-step protocol-grounded remediation advice for each flagged event. This directly reduces the manual documentation lookup burden that operators currently face during live incidents. NERC CIP compliance requires incident logging but not automated response; AYLA's explainable attribution provides an audit trail that supports compliance reporting while simultaneously assisting operators.

The cascaded per-scenario classifier (Section 6.4) provides a complementary fast path. By routing the majority of alerts through the cascade (sub-millisecond) and reserving the full AYLA pipeline for ambiguous events, mean response latency drops substantially without sacrificing attribution accuracy on high-confidence cases.

9 Conclusion

This paper presented Power Shield, implementing an end-to-end Automated Yield-Loss Alerting (AYLA) pipeline for power grid attack detection and LLM-guided remediation. Three principal results establish the pipeline's feasibility.

First, classifier feasibility: the SHAP-guided LightGBM/XGBoost stacked ensemble achieves AUC-ROC 0.9665, MCC 0.8020, and attack recall 94.6% at τ=0.560, surpassing the Hink et al. [10] combined-dataset baseline (90.56%) on the full heterogeneous MSU/ORNL dataset.

Second, LLM explainability proven statistically: AYLA achieves 100% RAG retrieval accuracy across all 13 protocol test cases and 90.5% correct protocol citation (181/200, 95% CI [85.6%, 93.8%]). A χ^2 goodness-of-fit test ($\chi^2 = 1931.42$, $p < 10^{-10}$) provides rigorous statistical proof that the LLM is genuinely grounding its responses in the retrieved protocol manuals—not selecting protocols randomly— demonstrating that the AYLA pipeline is both feasible and interpretably grounded.

Third, comparison with deployed systems: unlike current commercial ICS security platforms (Claroty, Dragos, Nozomi Networks), which provide passive network monitoring and alert surfacing but no automated remediation guidance, AYLA delivers step-by-step, protocol-grounded remediation advice for each flagged event—a capability absent from all known deployed ICS security systems.

An architecture comparison between the original single-LightGBM system (Run A, 84.5% LLM accuracy) and AYLA (Run B, 90.5%) shows a 6.0 pp improvement attributable to the ensemble classifier and cross-encoder reranking. The comparison also reveals a shift in failure character: Run A has concentrated, predictable failures (RTCI-01 always misattributed to R4-PM; FREQ-01 always fails), while AYLA has distributed failures (RTCI-01 is the only systematic failure mode).

The principal remaining limitation is RTCI-01 attribution ($0/7 = 0\%$ in AYLA), where phase-angle SHAP drivers suppress the log-channel vocabulary needed to distinguish RTCI-01 from relay-specific FDIA scenarios. Future work will address this via log-channel-aware RAG query construction, and integration of an online retraining loop to adapt the classifier to concept drift in adversarial grid telemetry.

References

[1] Gaoqi Liang, Junhua Zhao, Fengji Luo, Steven R. Weller, and Zhao Yang Dong. A review of false data injection attacks against modern power systems. *IEEE Transactions on Smart Grid*, 8(4):1630–1638, 2017.

[2] Yao Liu, Peng Ning, and Michael K. Reiter. False data injection attacks against state estimation in electric power grids. In *Proceedings of the 16th ACM Conference on Computer and Communications Security (CCS)*, pages 21–32. ACM, 2011.

[3] Chowdhury Monowar Hasan Ahmed, Venkata Reddy Palleti, and Aditya Mathur. WADI: A water distribution testbed for research in the design of secure cyber physical systems. In *Proceedings of the 3rd International Workshop on Cyber-Physical Systems for Smart Water Networks*, pages 25–28. ACM, 2017.

[4] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pages 4765–4774, 2017.

[5] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 9459–9474, 2020.

[6] Tommy Morris and Wei Gao. Industrial control system traffic data sets for intrusion detection research. In *Proceedings of the 8th International Conference on Critical Infrastructure Protection (ICCIP)*, pages 65–78. Springer, 2014.

[7] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pages 3146–3154, 2017.

[8] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. ACM, 2016.

[9] Manikant Panthi and Tanmoy Kanti Das. Intelligent intrusion detection scheme for smart power-grid using optimized ensemble learning on selected features. *International Journal of Critical Infrastructure Protection*, 39:100567, 2022.

[10] Raymond C. Borges Hink, Justin M. Beaver, Mark A. Buckner, Tommy Morris, Uttam Adhikari, and Shengyi Pan. Machine learning for power system disturbance and cyber-attack discrimination. In *2014 7th International Symposium on Resilient Control Systems (ISRCS)*, pages 1–8. IEEE, 2014.

[11] Hamad Naeem et al. Classification of intrusion cyber-attacks in smart power grids using deep ensemble learning with metaheuristic-based optimization. *Expert Systems*, 2025.

[12] Wenlong Xu et al. Graph attention and Kolmogorov–Arnold network based smart grids intrusion detection. *Scientific Reports*, 15, 2025.

[13] Tiago Craveiro et al. Detection of disturbances and cyber-attacks in smart grids using explainable machine learning. *Scientific Reports*, 2026.

[14] Scott M. Lundberg, Gabriel G. Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2:56–67, 2020.

[15] Mahboob Ur Rahman et al. Explainable AI for intrusion detection in industrial control systems, 2022.

[16] Mingyuan Li et al. IHGR-RAG: An enhanced retrieval-augmented generation framework for accurate and interpretable power equipment condition assessment. *Electronics*, 14(16):3284, 2025.

[17] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 24824–24837, 2022.

[18] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mistral 7B. *arXiv preprint arXiv:2310.06825*, 2023.